

Notes on Category

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0.1 Introduction

This is a note on category, facing to undergraduates, since our college don't even have one course on category! I don't know why, there is only one homological algebra.

This note contains definitely less content than a one term course. It will use basic undergraduate concepts as examples, introduce what is a category, how to describe using categories, why using categories, and eventually some basic category theories.

One should at least know:

- What is a group, group action, and group morphism
- Definition of topological space and continuous functions
- Tensor product of vector spaces

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Chapter 1

Inside a category

1.1 Basic Categories

1.1.1 Definition of categories

Category is basically a collection including mathematical objects and appropriate functions. For example:

- Category of groups and group morphisms.
- Category of topological spaces and continuous functions.
- Category of Sets and functions.
- Category of vector spaces over a fix field and linear transformations.

The reason that we define category is it unify different theorems and definitions. For example, direct product of groups/topological spaces/sets/vector spaces, turn out to be exactly the same thing from categorical view.

One wants to axiomize the notion of category. The basic version is:

Definition 1.1.1. *A category is a ordered pair (O, M) , where O is the collection of objects, and M is the collection of functions whose elements are called morphisms. The domain and codomain of a morphism must be an object. Additionally, for basic algebra we require:*

- For morphisms $f : X \rightarrow Y, g : Y \rightarrow Z$, their composition $g \circ f : X \rightarrow Z$ is inside M .
- Each object has the identity function $\text{id}_X : X \rightarrow X$ as a morphism.

Remark 1.1.1.1. *Technically, a collection is a set of sets. However, since there is no set of all sets, it's not legal to have a category of all sets.*

Set theory does provide solutions to this problem. **One way** is to weaken the used axioms: instead of using ZFC axioms, use less axioms and thus ZFC ensures there is a set, which can be pretended to be the "universe of all sets", and the fabricated universe satisfy the desired axioms. **The other solution** is to introduce stronger set axioms, which ensures that there exists a set, which can be a "universe of all sets" satisfying the ZFC axioms. Usually one choose to have infinitely many inaccessible cardinals for convenience.

However, the above definition is not sufficiently abstracted. Here is the actual definition, which is abstract.

Definition 1.1.2. *A category is $(O, H, \circ)$, where O is a collection of objects. For each pair of objects (X, Y) , H specifies a set $\text{Hom}(X, Y)$. For each (X, Y, Z) , $\circ$ specifies a function $\text{Hom}(X, Y) \times \text{Hom}(Y, Z) \rightarrow \text{Hom}(X, Z)$. The following axioms must be satisfied:*

- $\circ$ should be associative: $(f \circ g) \circ h = f \circ (g \circ h)$, in order to simulate the basic behaviour of functions.
- For each object X , there must be an element id_X , such that $\forall f : X \rightarrow Y, f \circ \text{id}_X = f$ and $\forall g : W \rightarrow X, \text{id}_x \circ g = g$. Obviously such element is unique: $\text{id}'_X = \text{id}'_X \circ \text{id}_X = \text{id}_X$.

Let's have a look at the power of categorical language - by restricting our language to appropriate objects and functions, we easily specifies what is an isomorphism:

Definition 1.1.3. *In a category, a morphism $f \in \text{Hom}(X, Y)$ is an isomorphism iff there is $g \in \text{Hom}(Y, X)$ such that $f \circ g = \text{id}_Y, g \circ f = \text{id}_X$. When such morphism exists, we say X and Y are isomorphic, write $X \cong Y$.*

Remark 1.1.1.2. *id_X is automatically isomorphism.*

Isomorphisms are exactly homeomorphisms in the category of topological spaces.

1.1.2 More examples of categories

Example 1.1.2.1. *Here are more daily-life categories.*

- *Category of Abelian groups and group morphisms.*
- *Given group G , category of all G -acting sets and functions satisfying $g \cdot f(x) = f(g \cdot x)$*
- *Category of rings and ring morphisms.*
- *Category of R -modules and R -linear maps where R is a commutative ring.*
- *Category of metric spaces and continuous functions.*
- *Category of metric spaces and isometry functions - functions $f : X \rightarrow Y$ such that $d_X(p, q) = d_Y(fp, fq)$.*
- *Category of partially ordered sets and order-preserving functions - $x \leq y \Rightarrow f(x) \leq f(y)$*
- *Category of normed spaces and continuous linear maps (functional analysis)*
- *Category of graphs and graph morphisms (Vertices mapped to vertices, Edge mapped to edges, preserving the end-point relations)*
- *Category of smooth manifolds and smooth functions.*
- *Category of topological groups and continuous group morphisms.*
- *Category of Lie groups and smooth group morphisms.*
- *Category of Lie algebra and Lie algebra morphisms.*

We also provide some toy categories, some of which appears to be useful.

Example 1.1.2.2. *This example is quite important.*

For a partially ordered set $(P, \leq)$, we abstractly define a category:

- *Objects: elements of P .*
- $Hom(X, Y) = \begin{cases} \emptyset, & X > Y \\ \{*\}, & X \leq Y \end{cases}$ where $\{*\}$ denotes a set with one element, and we don't care what the element is.
- *Since unique element presents in $Hom(X, Y)$, the composition of morphisms is automatically defined, and axioms automatically follows.*

One may consider $(\mathbb{R}, \leq)$, $([0, 1], \leq_{\mathbb{R}})$, $(\mathbb{N}, \leq)$, $(\mathbb{Z}, \leq)$ for concrete examples.

Example 1.1.2.3. For a set X , one may define a "discrete category" whose objects are set elements, and there are only identity morphisms.

Example 1.1.2.4. This example is quite important.

For a collection of sets $\mathcal{C}$, one may define the following category:

- *Objects: elements of $\mathcal{C}$.*

•

$$Hom(X, Y) = \begin{cases} \{\iota : X \rightarrow Y\}, & X \subseteq Y \\ \emptyset, & \text{otherwise} \end{cases}$$

Where ι is the inclusion map.

- *Since morphisms are functions, composition is automatically defined, and axioms automatically follows.*

Example 1.1.2.5. For a category C , one may define its opposite category C^{op} :

- *Objects: Objects of C*
- $Hom_{C^{op}}(X, Y) := Hom_C(X, Y)$, we use f^{op} to denote an element of $Hom_{C^{op}}(X, Y)$ where f is an element of $Hom_C(X, Y)$ to distinguish them.
- $f^{op} \circ g^{op} := (g \circ f)^{op}$ (Exercise to verify)

Example 1.1.2.6. One may also try defining product category: $P = C \times D$:

- *Objects: $Obj_P = Obj_C \times Obj_D$*

- *Homomorphisms:* $\text{Hom}_P((A, X), (B, Y)) := \text{Hom}_C(A, B) \times \text{Hom}_D(X, Y)$.
- $(f, \alpha) \circ_P (g, \beta) := (f \circ_C g, \alpha \circ_D \beta)$.

Example 1.1.2.7. (*Homotopy in topology*) For a topological space X , one may define a category:

- *Objects:* Point of X .
- *Morphisms:* Homotopy classes of paths connecting two points.
- *Composition:* Concatenation of two paths.

1.1.3 Language of categories

We introduce some basic definitions here.

Definition 1.1.4. A category where all morphisms are isomorphisms is called a groupoid. A groupoid with only one object is a group.

Definition 1.1.5. An automorphism of X is a $X \rightarrow X$ isomorphism. The automorphism group is formed by all automorphisms of X called $\text{Aut}(X)$ - not hard to verify that it is indeed a group.

Definition 1.1.6. An endomorphism of X is just a $X \rightarrow X$ morphism. Endomorphisms of an object automatically form a monoid, called $\text{End}(X)$.

Definition 1.1.7. Monomorphism imitate injections. A monomorphism in a category is a morphism $f : X \rightarrow Y$ such that:

$$\forall W, \forall g, g' : W \rightarrow X, (fg = fg') \Rightarrow (g = g')$$

Epimorphisms imitate surjections. An epimorphism in a category is a morphism $f : X \rightarrow Y$ such that:

$$\forall Z, \forall g, g' : Y \rightarrow Z, (gf = g'f) \Rightarrow g = g'$$

Example 1.1.3.1. Verify that in category of sets and functions, monomorphisms are exactly injections, epimorphisms are surjections.

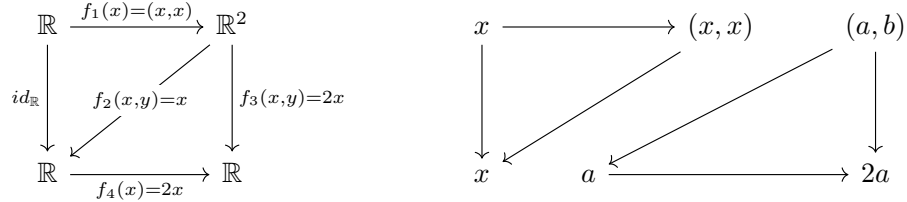
DANGER! Categories have very few axioms, it's usually impossible to deduce any non-trivial properties. For example, a monomorphism which is simultaneously an epimorphism may fail to become an isomorphism.

But on the other hand, categories do provide optional "axioms", from which a huge family of useful properties can be deduced. Though you still need to prove that your category satisfy the optional axioms, the axioms are in fact wise generalization of daily-life constructions, and you no longer need to prove a huge family of constructions, you only prove one general construction.

1.2 Universal property

1.2.1 Commutative diagram

In algebra, we frequently deal with a bunch of functions, which appear to be annoying and hard to handle. Clever Mathematicians organize them into a diagram, so that we can easily understand them. This is the commutative diagram.



The left hand side is a commutative diagram, illustrating the relation between functions. The right hand side is what you may write to verify if the diagram is commutative.

In a commutative diagram, there are nodes and arrows. Nodes present objects, and arrows present morphisms. Commutativity means, when a starting point and an end point is selected, no matter which path you pick, the path give the same morphism as a composition of morphisms on the path.

In the example, one may try to go from left-upper $\mathbb{R}$ to right-lower $\mathbb{R}$, and

"The diagram is commutative" is saying:

$$f_3 \circ f_1 = f_4 \circ id_{\mathbb{R}} = f_4 \circ f_2 \circ f_1$$

The three compositions corresponds to three possible paths. But this is not the only non-trivial equality, we also need $f_3 = f_4 \circ f_2$.

In order to verify if a diagram is commutative, one may try verifying each "minimal cycle", here you verify two triangles. In fact for complicated diagrams, it might turn out to be a complicated problem, especially when you are dealing with a non-planar graph.

Example 1.2.1.1. *THIS IS NOT COMMUTATIVE:*

$$\begin{array}{ccc}
 \mathbb{R} & \xrightarrow{f_1(x)=(x,x)} & \mathbb{R}^2 \\
 id_{\mathbb{R}} \downarrow & \swarrow f_2(x,y)=x & \downarrow f_3(x,y)=x+y \\
 \mathbb{R} & \xrightarrow{f_4(x)=2x} & \mathbb{R}
 \end{array}$$

Since the lower triangle is not commutative

There is an additional rule: the empty path is id_X where X is the node. This means if a cycling presents, the composition of morphisms should equal the identity function.

Example 1.2.1.2.

$$A \begin{array}{c} \xrightarrow{f} \\ \xleftarrow{g} \end{array} B$$

This means two functions are mutually inverse.

Example 1.2.1.3.

$$\begin{array}{ccc}
 \mathbb{R} & \xrightarrow{x^2+2x+2} & \mathbb{R} \\
 x+1 \updownarrow x-1 & & x+1 \updownarrow x-1 \\
 \mathbb{R} & \xrightarrow{x^2+2} & \mathbb{R}
 \end{array}$$

This is an example of conjugation. Sometimes, when the context is clear, one may write one bi-directional arrow to indicate an isomorphism which make the diagram commute in both way. But sometimes an isomorphism may fail to make the diagram commute in both way - see the next example:

Example 1.2.1.4.

$$\begin{array}{ccc} \mathbb{R} & \xrightarrow{id_{\mathbb{R}}} & \mathbb{R} \\ & \searrow \iota & \nearrow Re(z) \\ & \mathbb{C} & \end{array}$$

You cannot add the inverse arrow for $id_{\mathbb{R}}$, otherwise it would mean $\iota \circ id_{\mathbb{R}} \circ Re = id_{\mathbb{C}}$, which is absurd.

Here are some daily life examples:

Example 1.2.1.5.

$$\begin{array}{ccc} \mathbb{R} & \xrightarrow{x \mapsto xI} & \mathbb{R}^{3 \times 3} \\ & \searrow x \mapsto x^3 & \downarrow det \\ & & \mathbb{R} \end{array}$$

Example 1.2.1.6. In the category of sets and functions:

$$\begin{array}{ccccc} & & X & & \\ & f \nearrow & & \nwarrow \pi_X & \\ S & \xrightarrow{s \mapsto (fs, gs)} & X \times Y & & \\ & g \searrow & & \swarrow \pi_Y & \\ & & Y & & \end{array}$$

Where $\pi_X(x, y) = x, \pi_Y(x, y) = y$

Example 1.2.1.7. In definition of manifold, we require the transition map:

$$\begin{array}{ccc} U & \xleftarrow{\phi} & \Omega \subseteq \mathbb{R}^n \\ \uparrow \iota_U & & \downarrow \\ U \cap V & \begin{array}{c} \nwarrow \\ \nearrow \end{array} & \begin{array}{c} im(\phi \circ \iota_U) \\ \\ im(\psi \circ \iota_V) \end{array} \\ \uparrow \iota_V & & \downarrow \\ V & \xleftarrow{\psi} & \Gamma \subseteq \mathbb{R}^n \end{array}$$

This needs to be smooth

And also smooth maps are defined to be:

$$\begin{array}{ccc}
 M & \longrightarrow & N \\
 \updownarrow & & \updownarrow \\
 U & \longrightarrow & V \\
 \updownarrow & & \updownarrow \\
 \Omega \subseteq \mathbb{R}^m & \longrightarrow & \Gamma \subseteq \mathbb{R}^n \\
 \text{This needs to be smooth} & &
 \end{array}$$

1.2.2 Product

Definition 1.2.1. In a category, given a collection of objects $(X_i)_{i \in I}$, the product object is defined to be an object $P = \prod_{i \in I} X_i$ endowed with a collection of morphisms $(\pi_i : P \rightarrow X_i)_{i \in I}$, such that: $\forall W, \forall (f_i : W \rightarrow X_i)_{i \in I}, \exists! f : W \rightarrow P$, making the following diagram commute:

$$\begin{array}{ccccc}
 & & X_1 & & \\
 & f_1 \nearrow & & \nwarrow \pi_1 & \\
 S & \xrightarrow{f} & P & & \\
 & f_2 \searrow & & \swarrow \pi_2 & \\
 & & X_2 & & \\
 & f_i \searrow & & \swarrow \pi_i & \\
 & & \vdots & &
 \end{array}$$

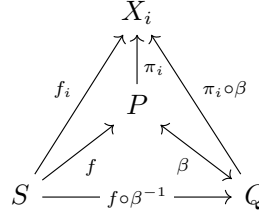
Where $\exists!$ reads as "uniquely exists", in case you didn't see this notation before.

Example 1.2.2.1. First let's understand this in the category of sets and functions, and focus on binary product. We've shown that $X \times Y$ with natural projections $\pi_X(x, y) = x, \pi_Y(x, y) = y$ and the construction would satisfy the above property:

$$\begin{array}{ccccc}
 & & X & & \\
 & f_X \nearrow & & \nwarrow \pi_X & \\
 S & \xrightarrow{f(s) := (f_X s, f_Y s)} & X \times Y & & \\
 & f_Y \searrow & & \swarrow \pi_Y & \\
 & & Y & &
 \end{array}$$

We need to verify if such function is unique. if g fit into the position, assume $g(s) = (a, b)$, then since $f_X = \pi_X \circ g$, we have $f_X(s) = \pi_X(g(s)) = \pi_X(a, b) = a$, thus $a = f_X(s)$. Similarly $b = f_Y(s)$, thus $g(s) = (f_X s, f_Y s)$. In conclusion, $g = f$.

Remark 1.2.2.2. Of course, the construction of product object is not unique - Any isomorphism with $P = \prod_{i \in I} X_i$, $\beta : Q \rightarrow P$ would give a different product object:

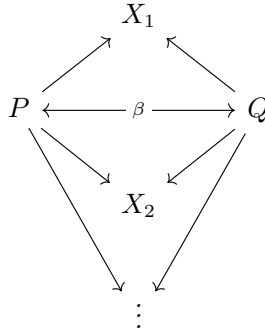


You can algebraically verify the commutativity involving X_i, S, Q , or deduce this commutativity from the commutativity of other three triangles.

Non-uniqueness of product object appears to be horrifying. However, we should verify that they are isomorphic together with the morphisms, which we should check in the next proposition.

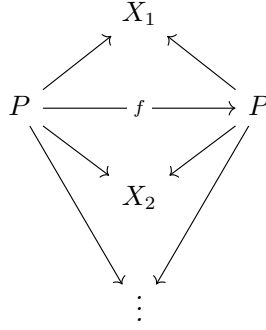
Proposition 1.2.2.3. One should always consider the product object with the projection morphisms, since the morphisms are the only linkage between P and X_i .

We claim that for two different product objects P, Q , there exists unique isomorphism $\beta : P \rightarrow Q$, making the diagram commute:



Well, how do we prove this? Obviously, by definition of Q being product, there exists a unique morphism $\beta : P \rightarrow Q$ making the diagram commute.

We need to find its inverse and prove that they are indeed inverse. The morphism $\alpha : Q \rightarrow P$ making the diagram commute uniquely exists by definition of P being a product. And we need to verify if α, β are inverse. It's easy. In the following diagram, since P on the right is a product, the following function f uniquely exists:



While both id_P and $\alpha \circ \beta$ can fit into the position of f , thus by uniqueness, $\text{id}_P = \alpha \circ \beta$. Similarly, $\beta \circ \alpha = \text{id}_Q$, thus α and β are mutually inverse.

Remark 1.2.2.4. The unique existence of the function f making the diagram including f_i, f, π_i, S, P, X_i is called "universal property of product".

Well, if you imagine that the category is the universe that you live in, then this property appears everywhere in the universe, so it's universal...

Remark 1.2.2.5. The definition of product object is merely a definition. Categories don't necessarily have product objects, and the product object don't need to be in fact a product. For example, in the category defined by usual ordering over $\mathbb{R}$, the product of $(x_i)_{i \in I}$ is merely $\inf\{x_i \mid i \in I\}$, which don't necessarily. (The morphisms between two objects in category defined by ordered set is unique, so no need to specify them)

Nevertheless, the product is still useful, easily characterizing the "product" of many Mathematical structures.

Example 1.2.2.6. Let's have a look at product object in the category of topological spaces and continuous functions, seeing how it characterises the topology on the cartesian product of sets.

Since all topological spaces can be viewed as sets, all continuous functions can be viewed as functions, there obviously exists a product set and projection

functions. We are interested in the topology of the product set P , in order to make both f and π_i continuous.

In order to make π_i continuous, we need to have pre-image of every open set being open. While in order to ensure f being continuous, P ought to have as few open sets as possible. This means the topology of P should be the topology with subbasis $\bigcup_{i \in I} \{\pi_i^{-1}(U) \mid U^{open} \subseteq X_i\}$ - Call this topology T_0 , any other finer topology T_1 would make $id_P : (P, T_0) \rightarrow (P, T_1)$ not continuous, while id_P is the only set function making relative diagrams commute.

If you fix the function f , this diagram is simply saying "If $\pi_i \circ f$ is continous, then f continuous". Or in other words, a function onto product space is continuous, if and only if it's continuous on each "dimension".

1.2.3 Initial and Final

We introduce these two definitions, which helps us establishing theories about universal properties.

Definition 1.2.2. In a category, an object I is called initial iff $\forall X, \exists! f : I \rightarrow X$.

Conversely, an object F is called final iff $\forall X, \exists f : X \rightarrow F$.

Example 1.2.3.1. .

- Inside the category of sets and functions, the final object is sets with exactly one element. If you know a bit set theory, the initial object is the empty set, and the function $f : \emptyset \rightarrow X$ obviously uniquely exists, called the empty function.
- Inside the category of k -Vector spaces and linear transformations, the final object and the initial object are both zero-dimensional spaces.
- Inside the category of groups, the final object and the initial object are both trivial groups.
- Inside the category of topolgical spaces, the final object is space with exactly one point, and the initial object is the empty space... Since topology is merely an augmentation of sets.

Proposition 1.2.3.2. *Initial objects mutually isomorphic. Similarly final objects are mutually isomorphic.*

证明. Let I_1, I_2 be initial objects. There exists a unique morphism $f : I_1 \rightarrow I_2$, and unique morphism $g : I_2 \rightarrow I_1$. Additionally, $I_1 \rightarrow I_1$ morphism is unique, which is id_{I_1} . Since $g \circ f$ is a $I_1 \rightarrow I_1$ morphism, thus $g \circ f = id_{I_1}$. Similarly, $f \circ g = id_{I_2}$, thus f, g are mutually inverse, and thus isomorphisms. $\square$

Example 1.2.3.3. *Inside the category defined by an ordered set, the initial element is a minimum element if exists. (The minimum element is the unique minimal element). Similarly, the final element is the unique maximum element if exists.*

Example 1.2.3.4. *Given a set X , define the following category:*

- *Object: Topologies on X .*
- *Morphism: $Hom(T_1, T_2) = \begin{cases} \{id_X\} & \text{if } id_X : (X, T_1) \rightarrow (X, T_2) \text{ continuous} \\ \emptyset & \text{otherwise} \end{cases}$*
- *Composition: composition of functions.*

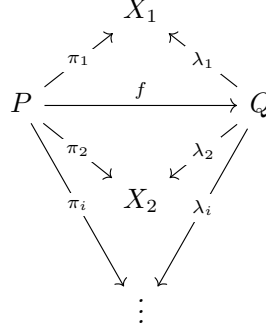
Easily verify that the category axioms are satisfied. The initial object is the discrete topology, and the final object is the indiscrete topology.

Example 1.2.3.5. *In the definition of product objects, we have a unique function to the product object making diagram commute. This means the product is somehow final.*

We can in fact define product to be the final object in the following categories. Let's say we want product object in category C . Give objects $(X_i)_{i \in I} \in Obj C$, define the following category:

- *Objects: $(P, (\pi_i)_{i \in I})$ where $P \in Obj C, \pi_i \in Hom_C(P, X_i)$*
- *Morphism: $Hom((P, \pi_i), (Q, \lambda_i)) = \{f \in Hom_C(P, Q) \mid \forall i, \pi_i = \lambda_i \circ f\}$*

$f\}$, which says f ought to make relative diagram commute:



- *Composition: Composition of functions*

Obviously $\text{id}_P \in \text{Hom}((P, \pi_i), (P, \pi_i))$, and axioms are satisfied. Note that id_P may not be an element of $\text{Hom}((P, \pi_i), (P, \lambda_i))$.

Product of X_i is the final object in this category.

1.2.4 Universal properties

Basically, universal property is a property that is universal... Usually the universal property is saying whenever you have a pile of objects and functions satisfying certain properties, you can find a unique function making relative diagrams commute.

on the other hand, universal property can be reworded as an initial/final object in a carefully defined category. This rewording makes proofs simpler, but not convenient in daily use.

Example 1.2.4.1. *The polynomial ring $R[x]$ is a ring together with ι_R, ι_x , where $\iota_R : R \rightarrow R[x]$ is a ring morphism and $\iota_x : \{x\} \rightarrow R[x]$ is a set function, such that whenever you have a ring morphism $\varphi : R \rightarrow S$ and a set-function $f : \{x\} \rightarrow S$, you have a ring morphism $\psi : R[x] \rightarrow S$, and the*

following diagram commute:

$$\begin{array}{ccc}
 & \{x\} & \\
 \iota_x \swarrow & & \searrow f \\
 R[x] & \xrightarrow{\psi} & S \\
 \iota_R \swarrow & & \searrow \varphi \\
 & R &
 \end{array}$$

In other words, $(R[x], \iota_R, \iota_x)$ is the initial object in the category of (S, φ, f) , where morphisms are ring morphisms $\phi : S_1 \rightarrow S_2$ making relative diagram commute.

Remark 1.2.4.2. The above definition involves two different categories: ring category and set category. This is not a big problem as long as you specifies how a ring is viewed as a set, how a ring morphism is viewed as a set function.

Example 1.2.4.3. We frequently have the notion of "generated subobject". Given a category C , in which all objects can be viewed as sets, all morphisms can be viewed as set-functions. For a object X and a subset $S \subseteq X$, the generated subobject is the initial object in the following category:

- Objects: Sub-object of X which contains S
- Morphisms: inclusion map $A \rightarrow B$ iff $A \subseteq B$.
- Composition: function composition.

Well, simply the smallest subobject containing S

Example 1.2.4.4. We also have free-generated notion. Given a category C , in which all objects can be viewed as sets, all morphisms can be viewed as set-functions. The free object generated by a set S is the initial object in the following category:

- Objects: (X, ι) where $X \in \text{Obj}C$ and $\iota : S \rightarrow X$ is a set function.
- Morphism: $\text{Hom}((X, \iota_1), (Y, \iota_2)) = \{f \in \text{Hom}_C(X, Y) \mid \iota_2 = f \circ \iota_1\}$
- Composition: inherited from C .

For example, free generated group, free R -modules, $|S|$ dimensional vector space, discrete topology, discrete graph...

Example 1.2.4.5. Given a category C , in which all objects can be viewed as sets, all morphisms can be viewed as set-functions. The quotient object defined by an equivalence relation, is the initial version of the following commutative diagram:

$$\begin{array}{ccc} \text{Set } X / \equiv & & \\ \uparrow p & \nwarrow q & \\ X & \xrightarrow{\pi} & Q \end{array}$$

Where p, q are set functions.

For example, quotient topological spaces must have the underlying set as a quotient set, being initial requires it has as much open sets as possible, while ensuring $\pi : X \rightarrow Q$ continuous.

For quotient group G/N , one may define the equivalence relation to be $g_1 \equiv g_2 \iff g_1 \in N \wedge g_2 \in N$. In order to have Q as a group, one must have the quotient group. Of course, if you define the equivalence relation to be $g_1 \equiv g_2 \iff g_1 N = g_2 N$ it directly gives the construction.

Example 1.2.4.6. This time we get back to things going on inside one category. We introduce the notion of coproduct.

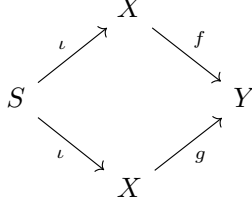
Recall that the dual category of C , C^{op} is the category with same objects, but arrows are reverted.

Coproduct is merely the product in the dual category. Coproduct also appears everywhere. We may also describe coproduct as an initial object in the following category:

- *Objects:* $(P, (\iota_i)_{i \in I})$ where $P \in \text{Obj } C, \iota_i : X_i \rightarrow P$.
- *Morphism:* morphisms between P_1, P_2 making relative diagrams commute.
- *Composition:* Inherits from C .

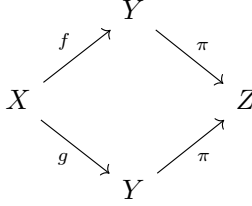
For example, the followings are coproducts in relative categories: Disjoint union of sets/topological spaces/graphs, free product of groups, direct sum of vector spaces and modules.

Definition 1.2.3. Given $f, g : X \rightarrow Y$, the equalizer defined to be the final $(S, \iota : S \rightarrow X)$ making the diagram commute:



In the category of sets, one realization of S, ι is $S := \{x \in X \mid f(x) = g(x)\}$, while ι is the inclusion map.

Definition 1.2.4. Given $f, g : X \rightarrow Y$, the coequalizer defined to be the initial (Z, π) making the diagram commute:



In the category of sets, one realization of Z, π is Z be the quotient set defined by the finest equivalence relation having $f(x) \equiv g(x)$, while π is the projection map. For example, $f, g : \mathbb{N} \rightarrow \mathbb{N}, f(n) = n, g(n) = n + 1$, then the coequalizer has unique element. If $f(n) = n, g(n) = 2n$, then the coequalizer can be viewed as set of odd numbers.

1.2.5 Categorical Limits

Example 1.2.5.1. Consider the following sequence of vector spaces. What's its "limit"?

$$\mathbb{R}, \mathbb{R}^2, \mathbb{R}^3, \dots$$

Obviously, it should be $\mathbb{R}^{\oplus \infty}$, consisting of eventually zero sequences of real numbers.

But what's its relation with $\mathbb{R}^n$? If we consider the process of expanding a vector space, with $\iota_n : \mathbb{R}^n \rightarrow \mathbb{R}^{n+1}, \iota(\vec{v}) = (\vec{v}, 0)$, rethink the sequence as an

infinite diagram:

$$\mathbb{R} \hookrightarrow \mathbb{R}^2 \hookrightarrow \mathbb{R}^3 \hookrightarrow \dots \hookrightarrow \mathbb{R}^n \hookrightarrow \mathbb{R}^{n+1} \hookrightarrow \dots$$

Then we can specify a inclusion map, explaining the relation between $\mathbb{R}^n$ and the limit:

$$j_n : \mathbb{R}^n \rightarrow \mathbb{R}^{\oplus \infty}, j_n(\vec{v}) = (v_1, v_2, \dots, v_n, 0, 0, 0, \dots)$$

Example 1.2.5.2. (*p*-adic integer) One classical construction of *p*-adic integers is to consider the limit of this diagram:

$$\mathbb{Z}/p\mathbb{Z} \longleftarrow \mathbb{Z}/p^2\mathbb{Z} \longleftarrow \mathbb{Z}/p^3\mathbb{Z} \longleftarrow \dots$$

$\mathbb{Z}_p$ the ring of *p*-adic integers can be presented by formal sums $\sum_{n=0}^{\infty} a_n p^n$ where $a_n \in \{0, 1, \dots, p-1\}$. It's the final set that has projection map onto every term in the sequece, making diagrams commute.

These two examples lead to the notion of categorical limits. However, currently, without the notion of functors, the definition is a bit fat...

Definition 1.2.5. In a category, you draw a diagram of objects and morphisms (not necessarily commute). The limit of the diagram is the final choice of object with a map π_i onto each object in the diagram, satisfying:

- For each pair of objects in the diagram X_i, X_j
- Let $M_{i,j}$ be the set of all $X_i \rightarrow X_j$ morphisms formed by composition of morphisms inside the diagram.
- We must have $\forall f \in M_{i,j}, f \circ \pi_i = \pi_j$

This is saying that though the diagram might be not commute, but "triangles" formed by π_i, π_j, f should commute.

Definition 1.2.6. Colimit is the limit in the dual category. You can also revert the arrows to define colimit - instead of a map from the limiting object to each object in the diagram, you have a map from objects in the diagram to the colimit object, making triangles commute.

Example 1.2.5.3. *The p-adic example is an example of limit, while the $\mathbb{R}^{\oplus\infty}$ example is an example of colimit. For the p-adic integer example, $\mathbb{Z}$ has a natural projection map onto each $\mathbb{Z}/p^n\mathbb{Z}$, but it's not final, it has an inclusion map into $\mathbb{Z}_p$.*

Remark 1.2.5.4. *The notion of limits and colimit follows from product and coproduct. One might have noticed that our definition is very general. Product can be viewed as a limit of diagrams without arrows, and coproduct can be viewed as its colimit. Meanwhile, equalizer can be viewed as limit, coequalizer can be viewed as colimit.*

The naming for these two concepts are chaotic. In daily life, people find colimit much more straight forward, and they call it direct limit $\varinjlim$, having a usual direction arrow; while limit is mind-twisting, so they call it inverse limit $\varprojlim$, having a reversed arrow.

*On the other hand, direct limits (colimits) usually have maps being injections, so they are also sometimes named as injective limit; while inverse limits (limit) usually have maps being projections, so they are called projective limits... **I won't suggest such naming**, we will see a common example violating this.*

I'll stick to limits and colimits in this article.

Example 1.2.5.5. *In a category defined from a partially ordered set, the limit is the infimum of selected objects (if the infimum uniquely exists), the colimit is the supremum if uniquely exists.*

Example 1.2.5.6. *In the category of sets, giving a totally sequence:*

$$\cdots \subseteq S_{-1} \subseteq S_0 \subseteq S_1 \subseteq \cdots$$

The diagram consists of these objects and inclusion maps, will have limit being the intersection, and colimit being the union.

Example 1.2.5.7. *(germ of functions) Given a manifold M and a point p , one would like to consider local smooth functions at p . A local function is a function defined on a neighborhood at p , $f : u_p \rightarrow \mathbb{R}$. Given an open set u , all $u \rightarrow \mathbb{R}$ smooth functions form a ring by pointwise addition and multiplication, call this ring R_u .*

Consider a nested local basis B at p , for example a family of open ball under a chart. We draw the diagram including $(R_u)_{u \in B}$, and for $u \subseteq v$, one have restriction map $\text{res} : R_v \rightarrow R_u$. Add the restriction maps into the diagram:

$$R_{B_{1/2}p} \xrightarrow{\text{res}} R_{B_{1/3}p} \xrightarrow{\text{res}} R_{B_{1/4}p} \longrightarrow \cdots$$

The colimit is the germ of functions, including equivalence classes of local functions. Two functions are equivalent iff they are equal on some neighborhood of p .

You see, though this is colimit, we have projection maps instead of inclusion map... I have no idea why they have the awkward naming.

Chapter 2

Functors

2.1 Functor and Review

The idea behind category is to discuss the relation between objects using morphisms. By properly define morphisms, discussions will be simplified. Well, obviously we do need relations between categories, which are called functors.

Definition 2.1.1. *A functor $\mathcal{F} : \mathcal{C} \rightarrow \mathcal{D}$ maps $\mathcal{C}$ -Objects to $\mathcal{D}$ -Objects, and also maps $f \in \text{Hom}_{\mathcal{C}}(X, Y)$ to $\mathcal{F}f \in \text{Hom}_{\mathcal{D}}(\mathcal{F}X, \mathcal{F}Y)$. Basic structure of category should be preserved - $\mathcal{F}(id_X) = id_{\mathcal{F}X}$, $\mathcal{F}(f \circ g) = \mathcal{F}f \circ \mathcal{F}g$.*

Remark 2.1.0.1. *It's necessary to check $\mathcal{F}id_X = id_{\mathcal{F}X}$, a counter example can be easily constructed.*

We introduce some standard notations:

- Use $\mathcal{C}, \mathcal{D}$ to denote categories, $\mathcal{F}, \mathcal{G}$ to denote functors.
- Use $\text{Obj}\mathcal{C}$ to denote collection of objects in $\mathcal{C}$.
- For $X, Y \in \text{Obj}\mathcal{C}$, $\text{Hom}_{\mathcal{C}}(X, Y)$ denotes the collection of homomorphisms between X, Y in the category $\mathcal{C}$. When there is no ambiguity, write $\text{Hom}(X, Y)$.
- $f : X \rightarrow Y$ means $f \in \text{Hom}(X, Y)$ in the category under discussion.

- $\prod_{i \in I} X_i$ denote the product object which is an object in the category, relative morphisms are usually denoted by π_i .
- $\coprod_{i \in I} X_i$ denote the coproduct.
- $\mathcal{Set}$ denote the category of sets and functions
- $\mathcal{Grp}$ denote the category of groups and group morphisms
- $\mathcal{Ring}$ denote the category of ring and ring morphisms.
- $\mathcal{Top}$ denote the category of topological spaces and continuous functions.
- $k - \mathcal{Vect}$ denote the category of vector spaces and linear transformations over field k .
- $R - \mathcal{Mod}$ denote the category of modules and linear maps over ring R .
- $G - \mathcal{Set}$ is the category of sets endowed with a G -group action, together with set functions preserving the group action.
- $\mathcal{C}^{op}$ denotes the opposite category of $\mathcal{C}$
- $\mathcal{C} \times \mathcal{D}$ denote the product category of $\mathcal{C}$ and $\mathcal{D}$
- $\mathcal{C} \coprod \mathcal{D}$ denote the coproduct category of $\mathcal{C}$ and $\mathcal{D}$ (Defined later)
- $\mathcal{C}^{\mathcal{D}}$ denote the category of $\mathcal{D} \rightarrow \mathcal{C}$ functors and natural transformations (Defined later)

Definition 2.1.2. *Categorical limit is formally defined for a functor $\mathcal{F} : I \rightarrow \mathcal{C}$, where I is a category that you may want to manually define.*

One may consider the following category, whose final object is the limit:

- *Object:* $(X, (\pi_i)_{i \in \text{Obj } I})$ where $X \in \text{Obj } \mathcal{C}$, $\pi_i \in \text{Hom}_{\mathcal{C}}(X, \mathcal{F}i)$
- *Morphism:* $\text{Hom}((X, (\pi_i)_{i \in \text{Obj } I}), (Y, (\lambda_i)_{i \in \text{Obj } I}))$ includes all $X \rightarrow Y$ morphisms such that $\forall i, j \in I, \forall \varphi \in \text{Hom}_I(i, j), \pi_j = \mathcal{F}\varphi \circ \pi_i$
- *Coposition:* Inherit from $\mathcal{C}$.

2.2 Functors and natural transformations

2.2.1 Functors and opposite categories

We recall the definition for opposite categories and product categories:

Definition 2.2.1. *The opposite category $\mathcal{C}^{op}$ has the followings:*

- *Object:* The same as $\mathcal{C}$. One may write X, X^{op} to distinguish objects of $\mathcal{C}$ and $\mathcal{C}^{op}$
- *Morphism:* $Hom_{\mathcal{C}^{op}}(X^{op}, Y^{op}) = Hom_{\mathcal{C}}(Y, X)$. Similarly one may write f, f^{op} .
- *Composition:* $(f^{op} \circ g^{op}) := (g \circ f)^{op}$.

Obviously $\mathcal{C} = (\mathcal{C}^{op})^{op}$.

Definition 2.2.2. $\mathcal{C} \times \mathcal{D}$ has:

- *Object:* $Obj(\mathcal{C} \times \mathcal{D}) := Obj\mathcal{C} \times Obj\mathcal{D}$
- *Morphism:* $Hom((A_{\mathcal{C}}, X_{\mathcal{D}}), (B_{\mathcal{C}}, Y_{\mathcal{D}})) := Hom_{\mathcal{C}}(A, B) \times Hom_{\mathcal{D}}(X, Y)$
- *Composition:* $(f, \varphi) \circ (g, \psi) := (f \circ_{\mathcal{C}} g, \varphi \circ_{\mathcal{D}} \psi)$

Here is an example:

Example 2.2.1.1. *For a category $\mathcal{C}$, $Hom_{\mathcal{C}}(-, -)$ is a $\mathcal{C}^{op} \times \mathcal{C} \rightarrow Set$ functor, where we use $-$ to denote where the variables are put. For $f : X' \rightarrow X, g : Y \rightarrow Y'$, the set function $Hom_{\mathcal{C}}(f, g) : Hom_{\mathcal{C}}(X, Y) \rightarrow Hom_{\mathcal{C}}(X', Y')$ is:*

$$Hom_{\mathcal{C}}(f, g)(\varphi) := g \circ \varphi \circ f$$

$$\begin{array}{ccc} X & \xrightarrow{\varphi} & Y \\ f \uparrow & & \downarrow g \\ X' & \xrightarrow{Hom_{\mathcal{C}}(f, g)(\varphi)} & Y' \end{array}$$

Of course, one never really writes $Hom_{\mathcal{C}}(f, g)(\varphi)$... This is toooo weird. Here I'm writing so since it's the basic example, plz don't write so in daily life.

Example 2.2.1.2. One $\text{Set}^{op} \rightarrow \text{Set}$ functor is defined to be:

- X mapped to $\mathcal{P}X$ the power set.
- $f : X \rightarrow Y$ mapped to the pre-image function $f^{-1} : \mathcal{P}Y \rightarrow \mathcal{P}X$.

Easy to verify that this is a functor.

Proposition 2.2.1.3. If $\mathcal{F} : \mathcal{C} \rightarrow \mathcal{D}$ is a functor, then obviously there is a functor $\mathcal{F}^{op} : \mathcal{C}^{op} \rightarrow \mathcal{D}^{op}$. In particular, if $\mathcal{G} : \mathcal{C}^{op} \rightarrow \mathcal{D}$, then $\mathcal{G}^{op} : \mathcal{C} \rightarrow \mathcal{D}^{op}$.

2.2.2 Natural transformation

In linear algebra, one learns that for finite dimensional vector space, the dual space V^* has the same dimension as V , and there is a natural injective morphism $V \rightarrow (V^*)^*$ - for finite dimensional case, the morphism is injective between two spaces of same dimensions, thus bijective.

People says " $(V^*)^*$ canonically isomorphic to V with respect to this morphism". Well, we have $\dim V = \dim V^*$, why this is not "canonical"?

Category theory provides a satisfying explanation to this. In daily life, instead of looking at a single element, one usually look at the global structure of the space, wondering whether two points in the space are somehow the same. Category theory suggests instead of looking at a single object, one may have a look at the whole category.

Example 2.2.2.1. When one suggest V *naturally isomorphic* to $(V^*)^*$, it means that one isomorphism can be chosen for each V , call it β_V , and whenever there is a morphism $\varphi : V \rightarrow W$, the corresponding morphism $(\varphi^*)^* : (V^*)^* \rightarrow (W^*)^*$ would make the following diagram commute:

$$\begin{array}{ccc} V & \xleftarrow{\beta_V} & (V^*)^* \\ \downarrow \varphi & & \downarrow (\varphi^*)^* \\ W & \xleftarrow{\beta_W} & (W^*)^* \end{array}$$

Example 2.2.2.2. Another example of natural isomorphism is $X \times Y \cong Y \times X$ in the category of sets.

In order to formalize, categorists came up with the following definition:

Definition 2.2.3. If $\mathcal{F}, \mathcal{G}$ are both $\mathcal{C} \rightarrow \mathcal{D}$ functors, a natural transformation η between them is an index collection of $\mathcal{D}$ morphisms, indexed by $\text{Obj}\mathcal{C}$, satisfying:

- For $X \in \text{Obj}\mathcal{C}$, $\eta_X \in \text{Hom}_{\mathcal{D}}(\mathcal{F}X, \mathcal{G}X)$
- It ought to make all following diagrams commute:

$$\begin{array}{ccccc}
 X & & \mathcal{F}X & \xrightarrow{\eta_X} & \mathcal{G}X \\
 \downarrow \varphi & & \downarrow \mathcal{F}\varphi & & \downarrow \mathcal{G}\varphi \\
 Y & & \mathcal{F}Y & \xrightarrow{\eta_Y} & \mathcal{G}Y
 \end{array}$$

In other words, $\forall \varphi \in \text{Hom}_{\mathcal{C}}(X, Y), \eta_Y \circ \mathcal{F}\varphi = \mathcal{G}\varphi \circ \eta_X$

One write $\eta : \mathcal{F} \rightsquigarrow \mathcal{G}$ to illustrate that η is a natural transformation from $\mathcal{F}$ to $\mathcal{G}$.

Definition 2.2.4. $\text{id}_{\mathcal{F}} : \mathcal{F} \rightsquigarrow \mathcal{F}$ is the natural transformation such that $(\text{id}_{\mathcal{F}})_X = \text{id}_{\mathcal{F}X}$. For functors $\mathcal{F}, \mathcal{G}, \mathcal{H} : \mathcal{C} \rightarrow \mathcal{D}$, two natural transformation $\eta : \mathcal{F} \rightsquigarrow \mathcal{G}, \zeta : \mathcal{G} \rightsquigarrow \mathcal{H}$ can compose into $\zeta \circ \eta : \mathcal{F} \rightsquigarrow \mathcal{H}$, by $(\zeta \circ \eta)_X = \zeta_X \circ \eta_X$, which can be verified to be a $\mathcal{F} \rightsquigarrow \mathcal{H}$ transformation.

Definition 2.2.5. The category $\mathcal{D}^{\mathcal{C}}$ has:

- Object: $\mathcal{C} \rightarrow \mathcal{D}$ functors.
- Morphisms: Natural transformation between functors.
- Composition: defined above.

Easy to verify that this is a category.

Definition 2.2.6. A natural isomorphism $\eta : \mathcal{F} \rightsquigarrow \mathcal{G}$ is a natural transformation such that all η_X are isomorphism. In other words, its an isomorphism inside the $\mathcal{D}^{\mathcal{C}}$ category - the inverse can be directly constructed by $\zeta_X := \eta_X^{-1}, \zeta : \mathcal{G} \rightsquigarrow \mathcal{F}$.

When one write down an isomorphism and claim that this is natural, one usually mean each side can be expressed as a functor, and the two functors are naturally isomorphic.

Example 2.2.2.3. One says in set category $X \times Y \cong^{\text{nat}} Y \times X$, this is saying the left hand side is a $\text{Set} \times \text{Set} \rightarrow \text{Set}$ functor, mapping X, Y to $X \times Y$, also mapping functions to suitable functions. Similarly, right hand side is another $\text{Set} \times \text{Set} \rightarrow \text{Set}$ functor, mapping X, Y to $Y \times X$. And a natural isomorphism is found between them.

Example 2.2.2.4. $V \cong (V^*)^*$ is a bit more complicated. First, the dual space is a $k - \text{Vect}^{\text{op}} \rightarrow k - \text{Vect}$ functor, mapping V to V^* , mapping $\varphi : V \rightarrow W$ to $\Phi : (W^*) \rightarrow (V^*)$, $\Phi(f) := f \circ \varphi$.

Thus the double dual is a $k - \text{Vect} \rightarrow k - \text{Vect}$ functor. In order to have the morphism defined in linear algebra being bijection, we restrict to the category of finite dimensional spaces. Thus both side are $k - \text{Vect} \rightarrow k - \text{Vect}$ functor - the left hand side is the identity functor, doing nothing.

We need some concrete exercise to establish solid understandings on natural transformations. Answers are after the problems.

Exercise 2.2.1. Fix a set S , the following is a functor $\mathcal{F} : \text{Set} \rightarrow \text{Set}$:

- $\mathcal{F}X := S \times X$
- $f : X \rightarrow Y, \mathcal{F}f(s, x) := (s, fx)$

Find natural transformation $\mathcal{F} \rightsquigarrow \text{id}_{\text{Set}}$ and $\text{id}_{\text{Set}} \rightsquigarrow \mathcal{F}$. Are there any natural isomorphisms?

Exercise 2.2.2. Define functor $\Delta : \text{Set} \rightarrow \text{Set}$:

- $\Delta X := X \times X$
- $\Delta f : X \times X \rightarrow Y \times Y, (x_1, x_2) \mapsto (fx_1, fx_2)$

Find a natural transformation $\text{id}_{\text{Set}} \rightsquigarrow \Delta$ and two natural transformations $\Delta \rightsquigarrow \text{id}_{\text{Set}}$

Exercise 2.2.3. Fix $A \in \text{Obj}\mathcal{C}$. Define functor $h_A : \mathcal{C} \rightarrow \text{Set}$:

- $h_A(X) := \text{Hom}_{\mathcal{C}}(A, X)$.
- $h_A f : \text{Hom}_{\mathcal{C}}(A, X) \rightarrow \text{Hom}_{\mathcal{C}}(A, Y), \varphi \mapsto f \circ \varphi$.

Now we have $\psi \in \text{Hom}_{\mathcal{C}}(B, A)$, construct a natural transformation $h_A \rightsquigarrow h_B$.

Exercise 2.2.4. Let $\mathcal{C}$ has initial object I . $\mathcal{F} : \mathcal{C} \rightarrow \mathcal{C}$ is defined to be:

- $\mathcal{F}X := I$
- $\mathcal{F}f := \text{id}_X$

Show that $\mathcal{F}$ is initial object in $\mathcal{C}^{\mathcal{C}}$.

Definition 2.2.7. By definition a functor $\mathcal{F} : \mathcal{C} \rightarrow \mathcal{D}$ specifies a set-function $\text{Hom}_{\mathcal{C}}(X, Y) \rightarrow \text{Hom}_{\mathcal{D}}(\mathcal{F}X, \mathcal{F}Y)$ for each pair of $\mathcal{C}$ objects. If all these set functions are injective, then the functor is called faithful. If all these set functions are surjective, then the functor is called full. A fully faithful functor is a full and faithful functor, making the set functions bijective.

Exercise 2.2.5. Recall that a groupoid is a category where all morphisms are isomorphisms, while a group can be viewed as a groupoid with exactly one object.

Let G be a groupoid with exactly one object. Show:

- A $G \rightarrow G$ functor can be viewed as a $G \rightarrow G$ group morphism, and vice versa.
- With the above correspondence, a group automorphism corresponds to a fully faithful functor.
- A group automorphism is inner automorphism, iff the corresponding functor has a natural isomorphism to the identity functor.

Exercise 2.2.6. Let $\varphi : G \rightarrow H$ be a group morphism. With this morphism, every set with H -action can be viewed as a set with G -action, defining a functor $\mathcal{F} : H\text{-Set} \rightarrow G\text{-Set}$. Explain this.

For an automorphism $\varphi : G \rightarrow G$ defining a functor $\mathcal{F} : G\text{-Set} \rightarrow G\text{-Set}$, show φ is inner automorphism iff $\exists \eta : \text{id}_{G\text{-Set}} \cong^{\text{nat}} \mathcal{F}$.

Answer 2.2.1. $\pi_X(s, x) := x$ is a natural transformation $\pi : \mathcal{F} \rightsquigarrow \text{id}_{\text{Set}}$, while for each $s \in S$, a natural transformation $(\iota_s)_X : X \rightarrow S \times X, x \mapsto (s, x)$ can be defined.

If $|S| = 1$, then the above transformations are mutually inverse. Otherwise take $X = \{*\}$, no bijection can be found between X and $S \times X$, so there can't be any isomorphisms.

Answer 2.2.2. $\eta_X : X \rightarrow \Delta X, x \mapsto (x, x)$ is obviously natural. Similarly, $(\pi_1)_X : \Delta X \rightarrow X, (a, b) \mapsto a$ is natural, so does $(\pi_2)_X : \Delta X \rightarrow X, (a, b) \mapsto b$.

Answer 2.2.3.

$$\eta : h_A \rightsquigarrow h_B, \eta_X : \text{Hom}_{\mathcal{C}}(A, X) \rightarrow \text{Hom}_{\mathcal{C}}(B, X), \varphi \mapsto \varphi \circ \psi$$

Answer 2.2.4. For each functor $\mathcal{G} : \mathcal{C} \rightarrow \mathcal{C}$, a unique natural transformation $\eta : \mathcal{F} \rightsquigarrow \mathcal{G}$ specifies the unique morphism from $I = \mathcal{F}X$ to $\mathcal{G}X$. By uniqueness of morphism, all relative diagrams must commute.

Answer 2.2.5. Trivial...

Answer 2.2.6. Let's use $(X, \cdot_G)$ to denote a G -set, where $\cdot_G$ is the group action.

- $\mathcal{F}(X, \cdot_H) := \mathcal{F}(X, \cdot_G)$ where $g \cdot_G x := \varphi(g) \cdot_H x$.
- For $f : (X, \cdot_H) \rightarrow (Y, \cdot_H)$, define $\mathcal{F}f(x) := f(x)$, which still preserves the group action:

$$\mathcal{F}f(g \cdot_G x) := f(\varphi(g) \cdot_H x) = \varphi(g) \cdot_H f(x) = g \cdot_G \mathcal{F}f(x)$$

- The above definition also preserves identity function and composition, thus the functor is clearly defined.

The second part is a bit twisting, since while X and $\mathcal{F}X$ appears to be the same set, they are in fact endowed with different group actions. The identity function may fail to preserve the group action.

$*$ denotes action on $\mathcal{F}X$, $\cdot$ denotes action on X

$$g * x := \varphi(g) \cdot x, \text{ while } \text{id}(g \cdot x) = g \cdot \text{id}(x) \neq g * \text{id}(x)$$

If φ is an inner automorphism, $\varphi(t) = g_0 t g_0^{-1}$, claim $\eta : \text{id}_{G\text{-Set}} \rightsquigarrow \mathcal{F}$, $\eta_X : (X, \cdot) \rightarrow (X, *)$, $\eta_X(x) := g_0 \cdot x$ is a natural transformation:

- Verify η_X is a morphism:

$$\eta_X(g \cdot x) = g_0 g \cdot x = g_0 g g_0^{-1} \cdot (g_0 \cdot x) = \varphi(g) \cdot \eta_X(x) = g * \eta_X(x)$$

- Verify the diagrams:

$$\begin{array}{ccc} X & \xrightarrow{\eta_X} & \mathcal{F}X \\ \downarrow f & & \downarrow \mathcal{F}f \\ Y & \xrightarrow{\eta_Y} & \mathcal{F}Y \end{array} \quad \begin{array}{ccc} x & \xrightarrow{\quad} & g_0 \cdot x \\ \downarrow & & \downarrow \\ f(x) & \xrightarrow{\quad} & g_0 \cdot f(x) \end{array} \quad \begin{array}{ccc} & & f(g_0 \cdot x) \\ & \searrow \text{???} & \\ & = & \end{array}$$

$$f(g_0 \cdot x) = f(g_0 g_0 g_0^{-1} \cdot x) = f(g_0 * x) = g_0 * f(x) = g_0 g_0 g_0^{-1} \cdot f(x) = g_0 \cdot f(x)$$

- Obviously η_X is an isomorphism in $G\text{-Set}$, thus a natural isomorphism between $\text{id}_{G\text{-Set}}$ and $\mathcal{F}$.

If there exists a natural isomorphism $\eta : \text{id}_{G\text{-Set}} \cong \mathcal{F}$, then consider G itself as a $G\text{-Set}$ by left multiplication, say the action is $\cdot$, and action on $\mathcal{F}G$ is $*$. By definition, $g * x = \varphi(g) \cdot x$.

The key here is Right multiplication by $h \in G$ is a $G \rightarrow G$ morphism, call this morphism r_h . Let $g_0 = \eta_G(e)$, since $\mathcal{F}r_h = r_h$ by definition of the functor, we have:

$$\mathcal{F}r_h \circ \eta_G = \eta_G \circ r_h$$

And define $g_0 := \eta_G e$

$$(r_h \circ \eta_G)(e) = (\eta_G \circ r_h)(e)$$

$$g_0 h = \eta_G(h)$$

The equality holds $\forall h \in G$, thus η_G is simply left multiplication by g_0 . On the other hand, since η_G is a morphism:

$$\eta_G(h) = \eta_G(h \cdot e) = h * \eta_G(e) = \varphi(h)g_0$$

$$\varphi(h)g_0 = g_0 h$$

$$\varphi(h) = g_0 h g_0^{-1}$$

2.2.3 Yoneda Lemma

There is one theorem about natural transformations. For the moment, one may just do its proof as an exercise. I don't really understand how to use it either.

Theorem 2.2.1. *Let $h_A : \mathcal{C} \rightarrow \mathcal{Set}$ be the functor $h_A(X) := \text{Hom}_{\mathcal{C}}(A, X)$. Given a functor $\mathcal{F} : \mathcal{C} \rightarrow \mathcal{Set}$, we have a natural isomorphism:*

$$\text{Nat}(h_A, \mathcal{F}) \cong \mathcal{F}(A)$$

Where Nat denote the set of natural transformations between the two given functors, while $\mathcal{F}A$ is a set.

证明. We need to first specify what are the functors on two sides. They are both $\mathcal{Set}^{\mathcal{C}} \times \mathcal{C}^{op} \rightarrow \mathcal{Set}$ functors, where $\mathcal{F} \in \text{Obj}(\mathcal{Set}^{\mathcal{C}})$, $A \in \text{Obj}\mathcal{C}$. It's easy to verify that it's $\mathcal{C}^{op}$ if you try write down the morphisms.

Let's first specify the isomorphism for a fixed pair of $A, \mathcal{F}$. There is a unique way to put everything together and get something of correct type:

$$\Phi : \text{Nat}(h_A, \mathcal{F}) \rightarrow \mathcal{F}(A), \quad \Phi(\eta) := \eta_A(id_A)$$

Where η_A is a set function $\text{Hom}(A, A) \rightarrow \mathcal{F}A$, and $id_A \in \text{Hom}(A, A)$.

$$\Psi : \mathcal{F}A \rightarrow \text{Nat}(h_A, \mathcal{F}), \quad (\Psi(x)^{\in \mathcal{F}A})_B[f : A \rightarrow B] := \mathcal{F}f(x)$$

Need to verify if they are mutually inverse:

$$\Phi\Psi(x) = (\Psi x)_A(id_A) = \mathcal{F}(id_A)(x) = id_{\mathcal{F}A}x = x$$

$$\Psi\Phi(\eta)[f : A \rightarrow B] = \mathcal{F}f(\Phi\eta) = \mathcal{F}f(\eta_A(id_A))$$

Since η is natural transformation, and $h_A f : \text{Hom}(A, A) \rightarrow \text{Hom}(A, B)$, $g \mapsto f \circ g$ by definition, we have:

$$= (\mathcal{F}f \circ \eta)(id_A) = (\eta \circ h_A f)(id_A) = \eta(h_A f(id_A)) := \eta(f \circ id_A) = \eta f$$

Eventually we verify if these two morphisms are natural. First, assume

there is a morphism $f : B \rightarrow A$, and we want to verify the diagram:

$$\begin{array}{ccccc}
 A & & \text{Nat}(h_A, \mathcal{F}) & \longleftrightarrow & \mathcal{F}A & & \eta & \longleftrightarrow & \eta_A(id_A) \\
 \uparrow f & & \downarrow ??? & & \downarrow \mathcal{F}f & & \downarrow & & \downarrow \\
 B & & \text{Nat}(h_B, \mathcal{F}) & \longleftrightarrow & \mathcal{F}B & & \zeta & \longleftrightarrow & \zeta_B(id_B) \text{ vs } \mathcal{F}f(\eta_A(id_A))
 \end{array}$$

$$\zeta_X : h_B(X) \rightarrow \mathcal{F}X, \quad \zeta_X(g : B \rightarrow X) := \eta_X(g \circ f)$$

The commutativity is easily verified:

$$\zeta_B(id_B) := \eta_B(id_B \circ f) = \eta_B(f)$$

$$\mathcal{F}f(\eta_A(id_A)) = (\mathcal{F}f \circ \eta)(id_A) = (\eta \circ h_A f)(id_A) = \eta f$$

Thus commutative.

We still need to verify if there is a natural transformation $\gamma : \mathcal{F} \rightarrow \mathcal{G}$, relative diagrams commute:

$$\begin{array}{ccccc}
 \mathcal{F} & & \text{Nat}(h_A, \mathcal{F}) & \longleftrightarrow & \mathcal{F}A & & \eta & \longleftrightarrow & \eta(id_A) \\
 \downarrow \gamma & & \downarrow ??? & & \downarrow \gamma_A & & \downarrow & & \downarrow \\
 \mathcal{G} & & \text{Nat}(h_A, \mathcal{G}) & \longleftrightarrow & \mathcal{G}A & & \zeta & \longleftrightarrow & \zeta(id_A) \text{ vs } \gamma_A(\eta(id_A))
 \end{array}$$

$$\zeta_X : h_A X \rightarrow \mathcal{G}X, \quad \zeta_X(f : A \rightarrow X) := \gamma_X \eta_X(f)$$

The commutativity trivially follows:

$$\zeta(id_A) = \gamma_A \eta_A(id_A)$$

In conclusion, the isomorphism is natural. $\square$

2.2.4 Adjoint Functor

2.3 Limit and Adjoint functors

2.3.1 Comma Category

2.3.2 Completeness and completion

2.3.3 Adjoint and limit

Chapter 3

More on categories

3.1 Revision and examples

3.2 Closed Monoidal categories

3.2.1 Monoidal category

3.2.2 Extranatural transformation

3.2.3 Closed category

3.2.4 Examples of closed monoidal categories

3.2.5 Enriched categories

3.3 Chain complex and exact sequences

3.3.1 Zero object and zero morphism

3.3.2 Abelian category